FINITENESS OF THE HIGHER DIRECT IMAGES OF A COHERENT MODULE ON THE COMPLEMENT OF A CLOSED SUBSET, TESTED ON LOCAL COHOMOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-local-cohomology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-local-cohomology/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

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Allowed standard prerequisites (editorial).

- The extension of a coherent module on the complement of a closed subset to a coherent module on the ambient scheme. Invoked at the start of the counted proof (source lines 1749-1751) as a cross-chapter pointer. Produces the finite module whose local cohomology controls the higher direct images.
- The identification of the local cohomology of a finite module with the cohomology of the complement, including the exact sequence and the degree shifts. Invoked inside the counted proof (source lines 1758-1766) as pointers to a lemma of this chapter and two sections of another chapter. Converts the question about higher direct images into a question about local cohomology of the extended module.

Source reference. Original reference properties-lemma-lift-finite-presentation: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1749–1749. Navigation pointer only: the counted text refers to the Stacks item properties-lemma-lift-finite-presentation, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference coherent-lemma-quasi-coherence-higher-direct-images: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1756–1756. Navigation pointer only: the counted text refers to the Stacks item coherent-lemma-quasi-coherence-higher-direct-images, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own

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Source reference. Original lemma lemma-local-cohomology-is-local-cohomology: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 405–405. Navigation pointer only: the counted text refers to the Stacks item lemma-local-cohomology-is-local-cohomology, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference cohomology-section-cohomology-support: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1760–1760. Navigation pointer only: the counted text refers to the Stacks item cohomology-section-cohomology-support, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference cohomology-section-cohomology-support-bis: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 289–289. Navigation pointer only: the counted text refers to the Stacks item cohomology-section-cohomology-support-bis, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference coherent-lemma-quasi-coherent-affine-cohomology-zero: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 397–397. Navigation pointer only: the counted text refers to the Stacks item coherent-lemma-quasi-coherent-affine-cohomology-zero, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-local-cohomology: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 562–562. Navigation pointer only: the counted text refers to the Stacks item lemma-local-cohomology, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let A be a Noetherian ring and let $I \subset A$ be an ideal. Set $X = \mathrm{Spec}(A)$, $Z = V(I)$, $U = X \setminus Z$, and $j : U \to X$ the inclusion morphism. Let $\mathcal{F}$ be a coherent $\mathcal{O}_U$-module. Then*

1. *there exists a finite A-module M such that $\mathcal{F}$ is the restriction of $\widetilde{M}$ to U,*
2. *given M there is an exact sequence*
$$0 \to H^0_Z(M) \to M \to H^0(U, \mathcal{F}) \to H^1_Z(M) \to 0$$
and isomorphisms $H^p(U, \mathcal{F}) = H^{p+1}_Z(M)$ for $p \geq 1$,
3. *given M and $p \geq 0$ the following are equivalent*
   (a) *$R^p j_* \mathcal{F}$ is coherent,*
   (b) *$H^p(U, \mathcal{F})$ is a finite A-module,*
   (c) *$H^{p+1}_Z(M)$ is a finite A-module,*
4. *if the equivalent conditions in (3) hold for $p = 0$, we may take $M = \Gamma(U, \mathcal{F})$ in which case we have $H^0_Z(M) = H^1_Z(M) = 0$.*

Proof. By Properties, Lemma properties-lemma-lift-finite-presentation there exists a coherent $\mathcal{O}_X$-module $\mathcal{F}'$ whose restriction to U is isomorphic to $\mathcal{F}$. Say $\mathcal{F}'$ corresponds to the finite A-module M as in (1). Note that $R^p j_*\mathcal{F}$ is quasi-coherent (Cohomology of Schemes, Lemma coherent-lemma-quasi-coherence-higher-direct-images) and corresponds to the A-module $H^p(U, \mathcal{F})$. By Lemma lemma-local-cohomology-is-local-cohomology and the discussion in Cohomology, Sections cohomology-section-cohomology-support and cohomology-section-cohomology-support-bis we obtain an exact sequence

$$0 \to H^0_Z(M) \to M \to H^0(U, \mathcal{F}) \to H^1_Z(M) \to 0$$

and isomorphisms $H^p(U, \mathcal{F}) = H^{p+1}_Z(M)$ for $p \geq 1$. Here we use that $H^j(X, \mathcal{F}') = 0$ for $j > 0$ as X is affine and $\mathcal{F}'$ is quasi-coherent (Cohomology of Schemes, Lemma coherent-lemma-quasi-coherent-affine-cohomology-zero). This proves (2). Parts (3) and (4) are straightforward from (2); see also Lemma lemma-local-cohomology. $\square$